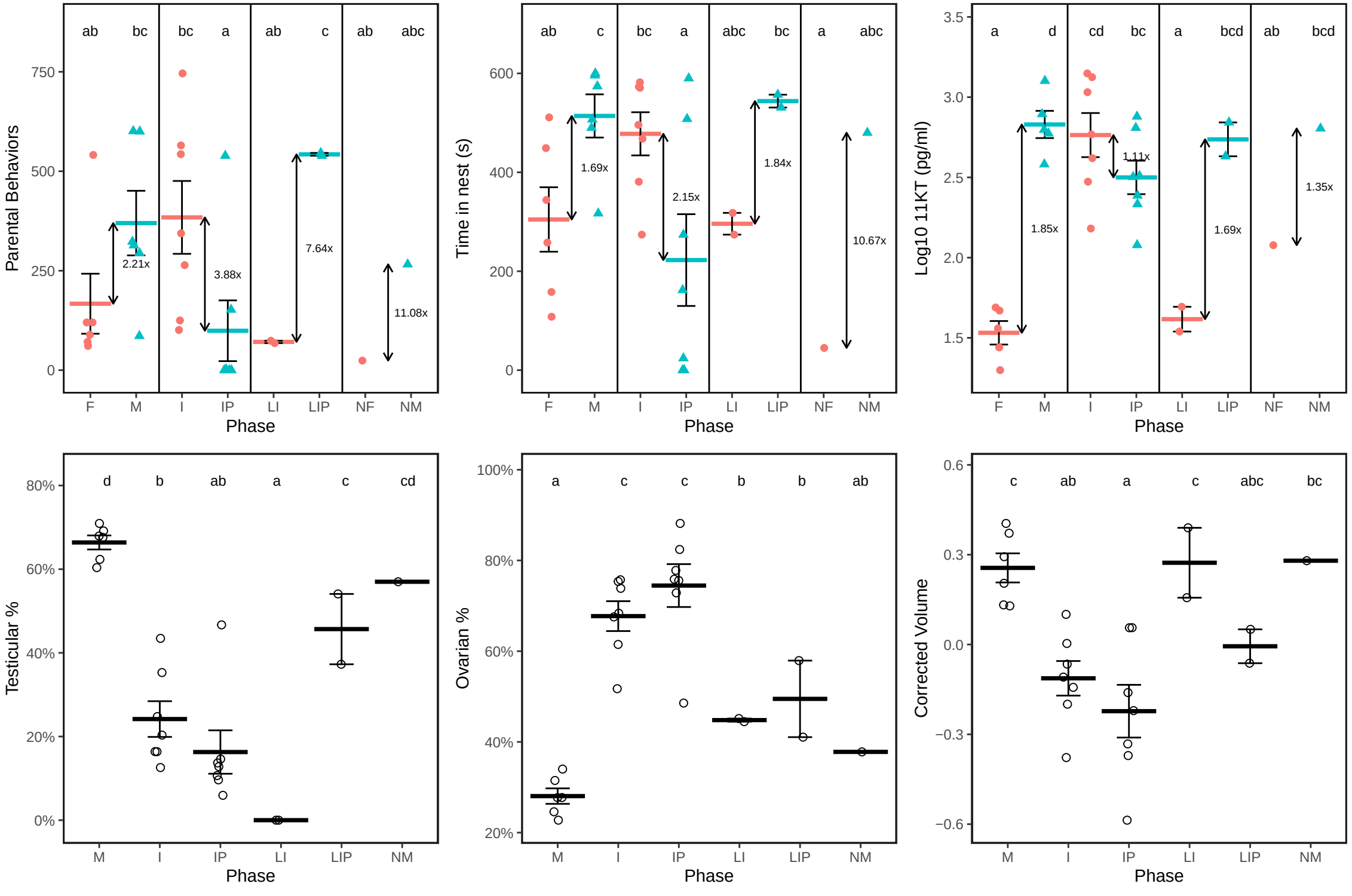
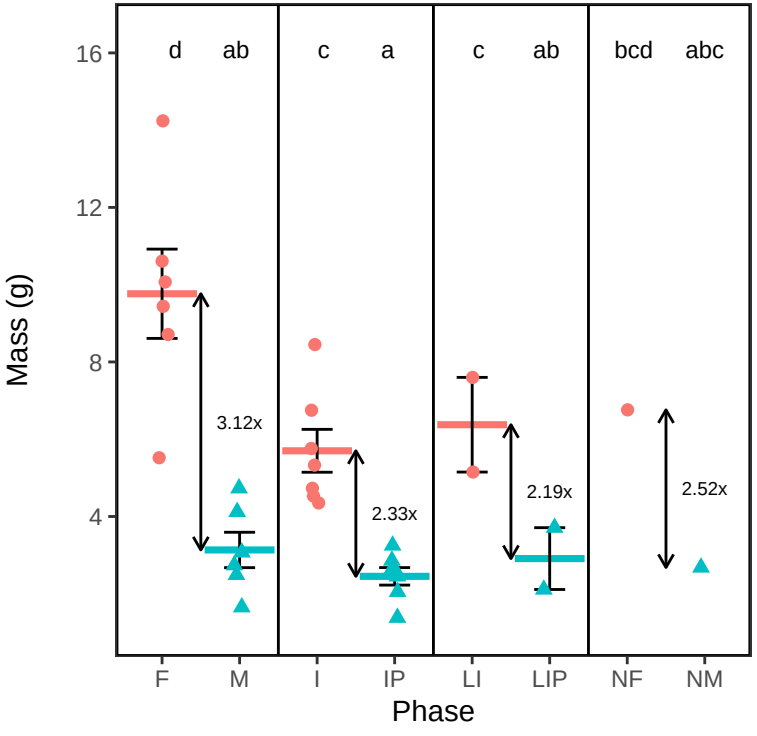
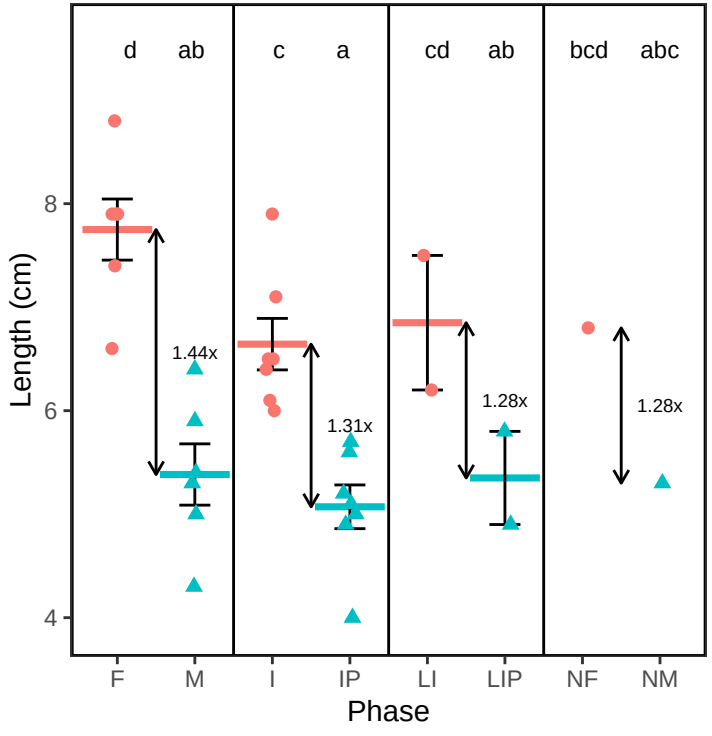
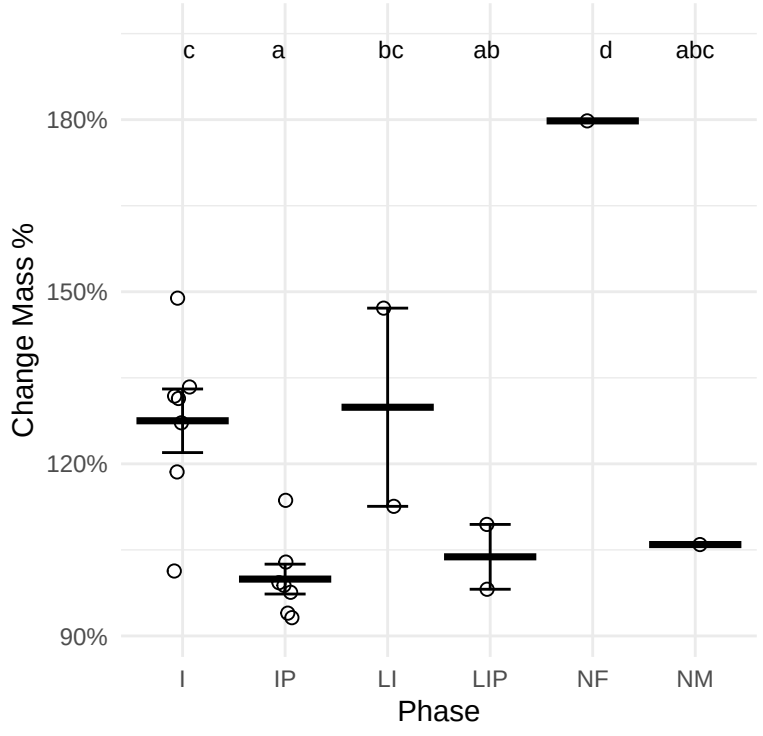
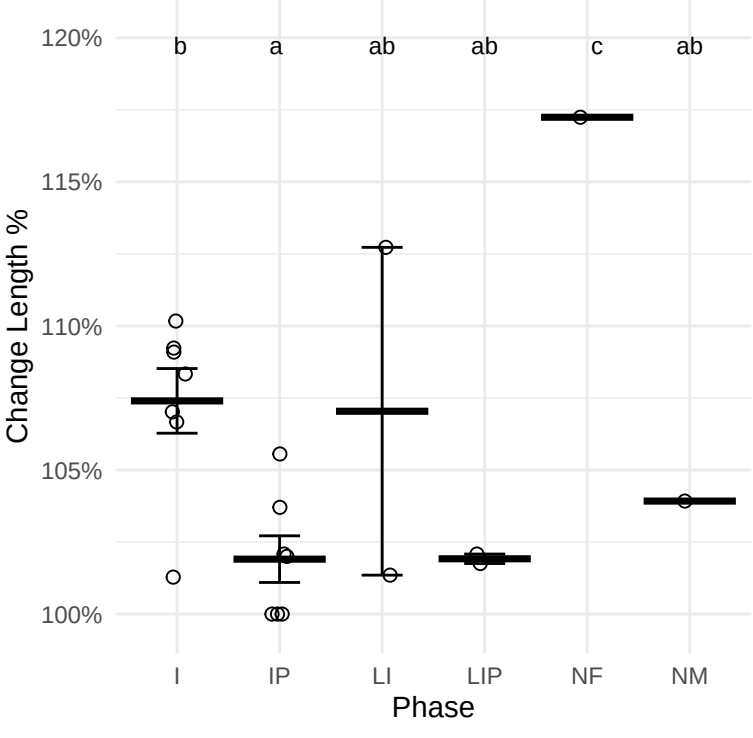
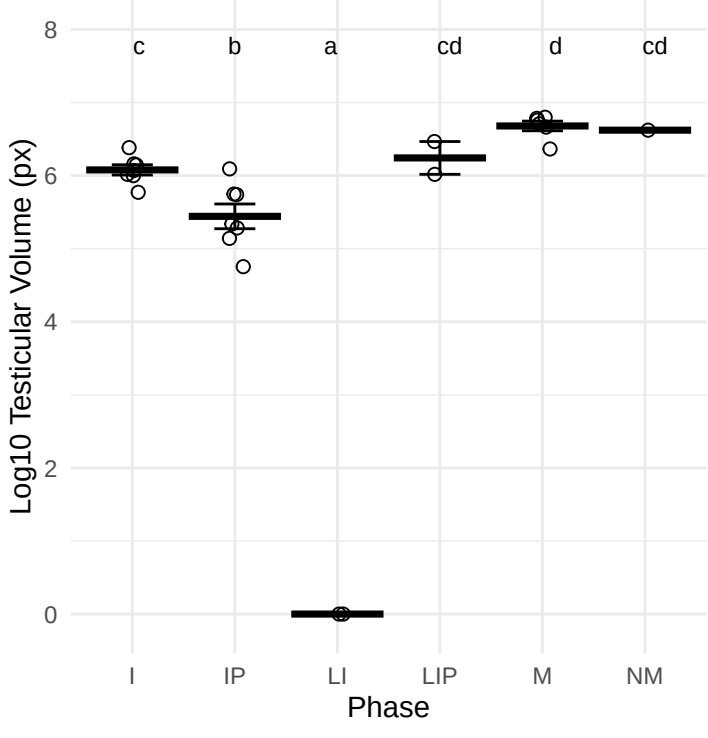

Parental Behaviors
Time in nest (s)
Log10 11KT (pg/ml)
Testicular %
Ovarian %
Corrected Volume
Phase
F
M
I
IP
LI
LIP
NF
NM
2.21x
3.88x
7.64x
11.08x
1.69x
2.15x
1.84x
10.67x
1.85x
1.11x
1.69x
1.35x

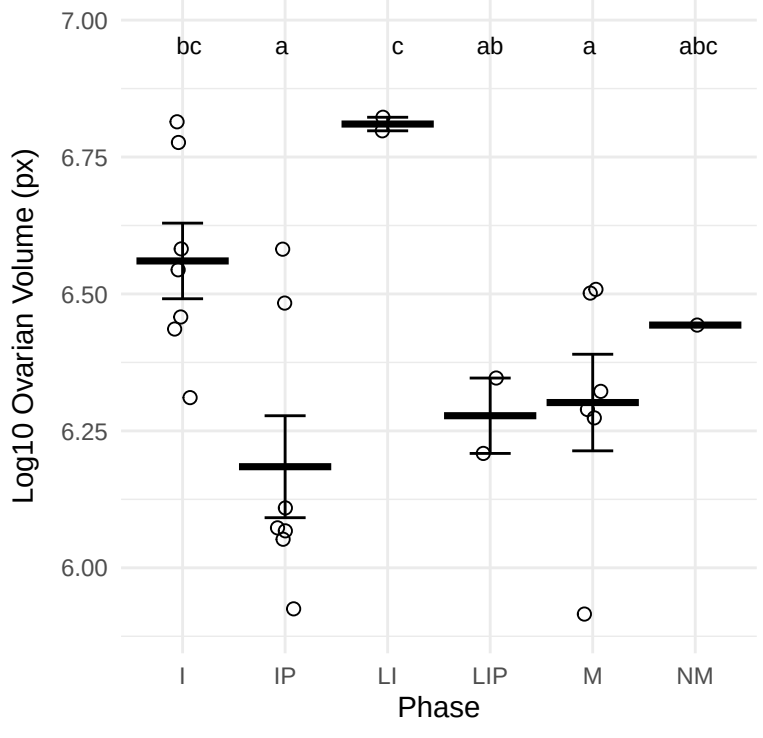

Mass (g)
16
12
8
4
d
ab
c
a
c
ab
bcd
abc
3.12x
2.33x
2.19x
2.52x
F
M
I
IP
LI
LIP
NF
NM
Phase

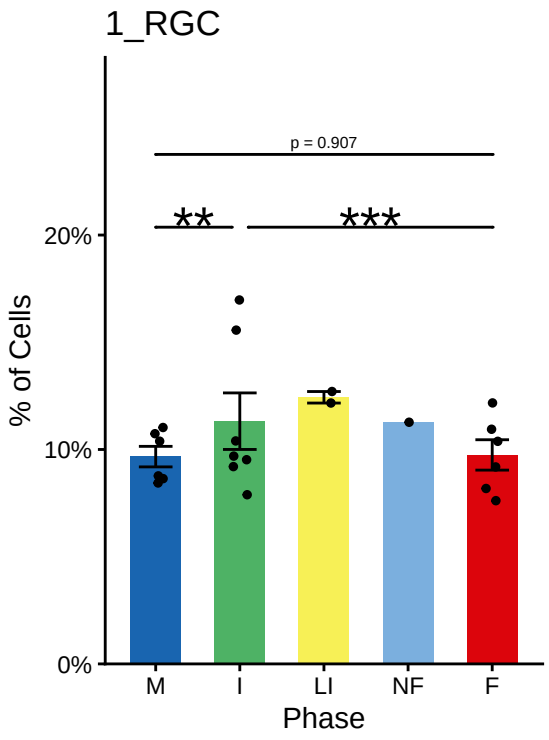

Length (cm)
8
6
4
d
ab
c
a
cd
ab
bcd
abc
1.44x
1.31x
1.28x
1.28x
F
M
I
IP
LI
LIP
NF
NM
Phase

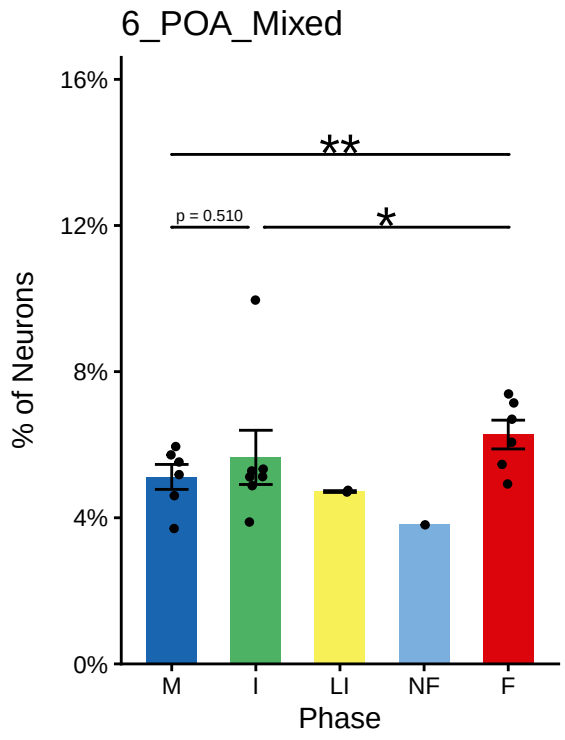

Change Mass %
180%
150%
120%
90%
c
a
bc
ab
d
abc
I
IP
LI
LIP
NF
NM
Phase

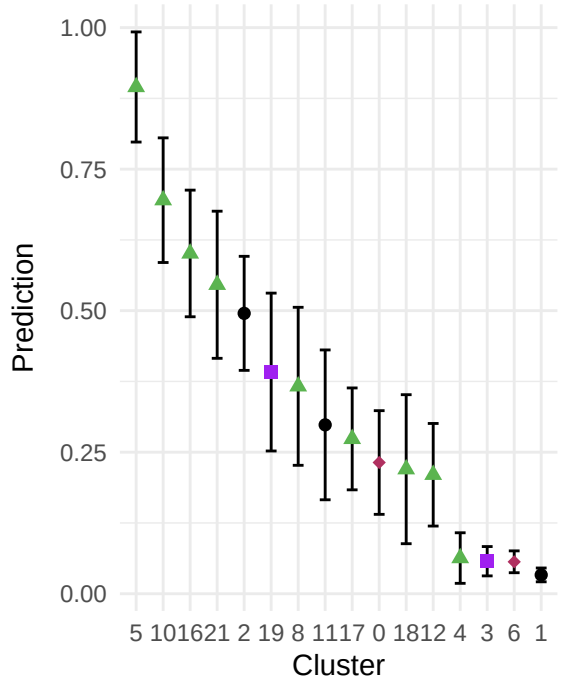

Change Length %
120%
115%
110%
105%
100%
b
a
ab
ab
c
ab
I
IP
LI
LIP
NF
NM
Phase

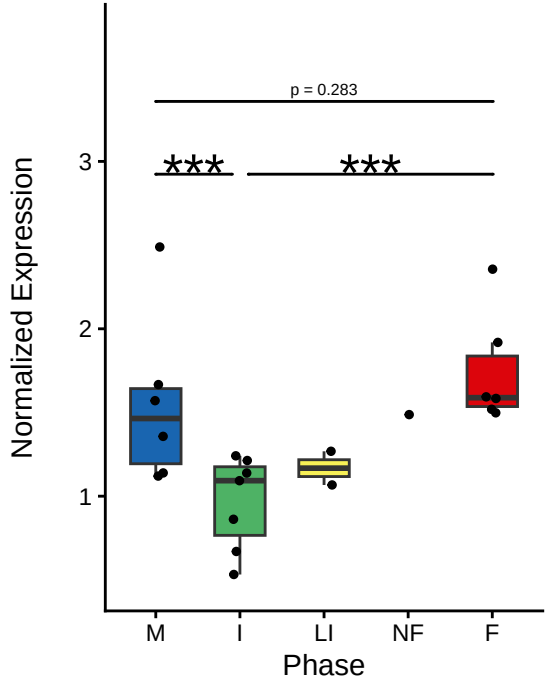

Log10 Testicular Volume (px)
8
6
4
2
0
c
b
a
cd
d
cd
I
IP
LI
LIP
M
NM
Phase

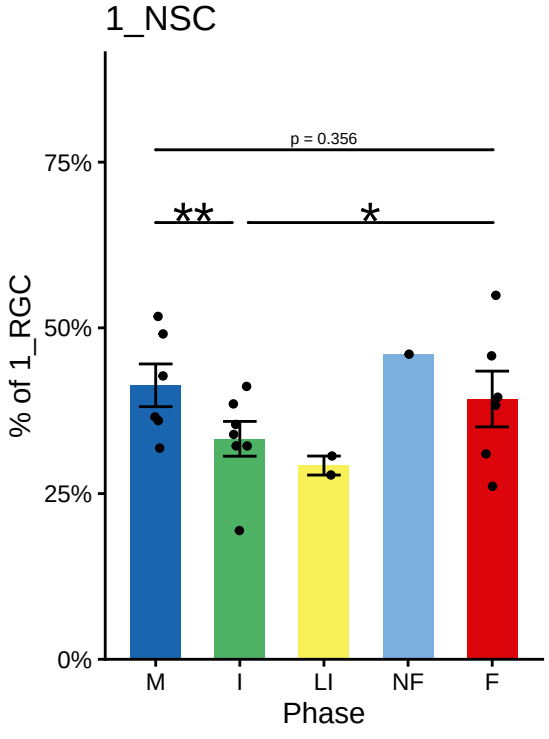

Log10 Ovarian Volume (px)
7.00
6.75
6.50
6.25
6.00
bc
a
c
ab
a
abc
I
IP
LI
LIP
M
NM
Phase

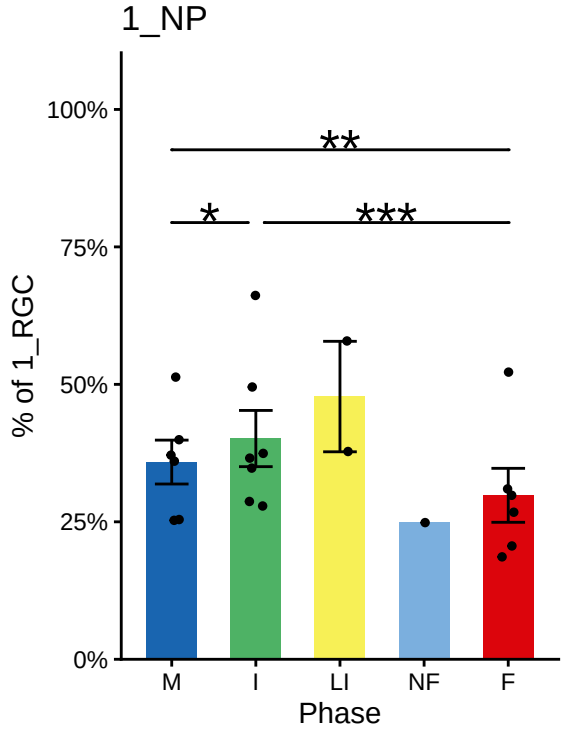

1_RGC
p = 0.907
**
***
% of Cells
20%
10%
0%
M
I
LI
NF
F
Phase

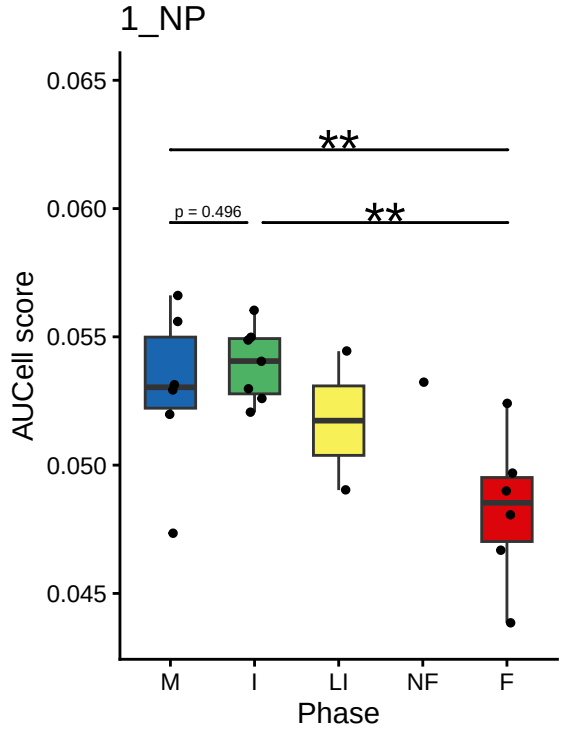

6_POA_Mixed
**
p = 0.510
*
% of Neurons
16%
12%
8%
4%
0%
M
I
LI
NF
F
Phase

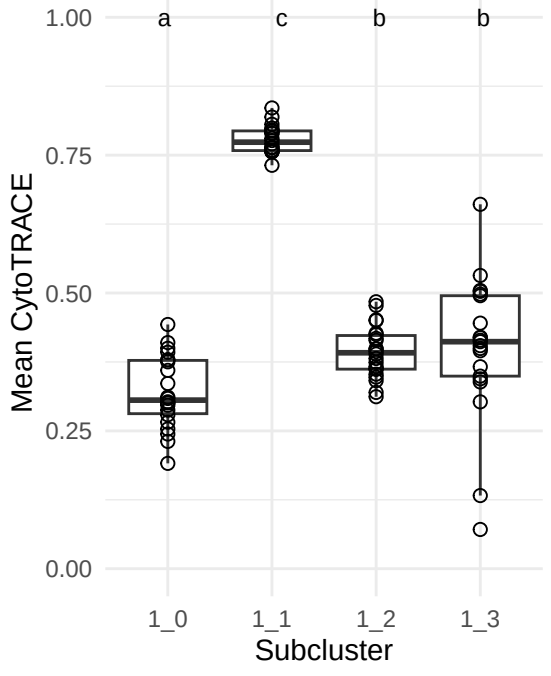

Prediction
1.00
0.75
0.50
0.25
0.00
5 10 16 21 2 19 8 11 17 0 18 12 4 3 6 1
Cluster

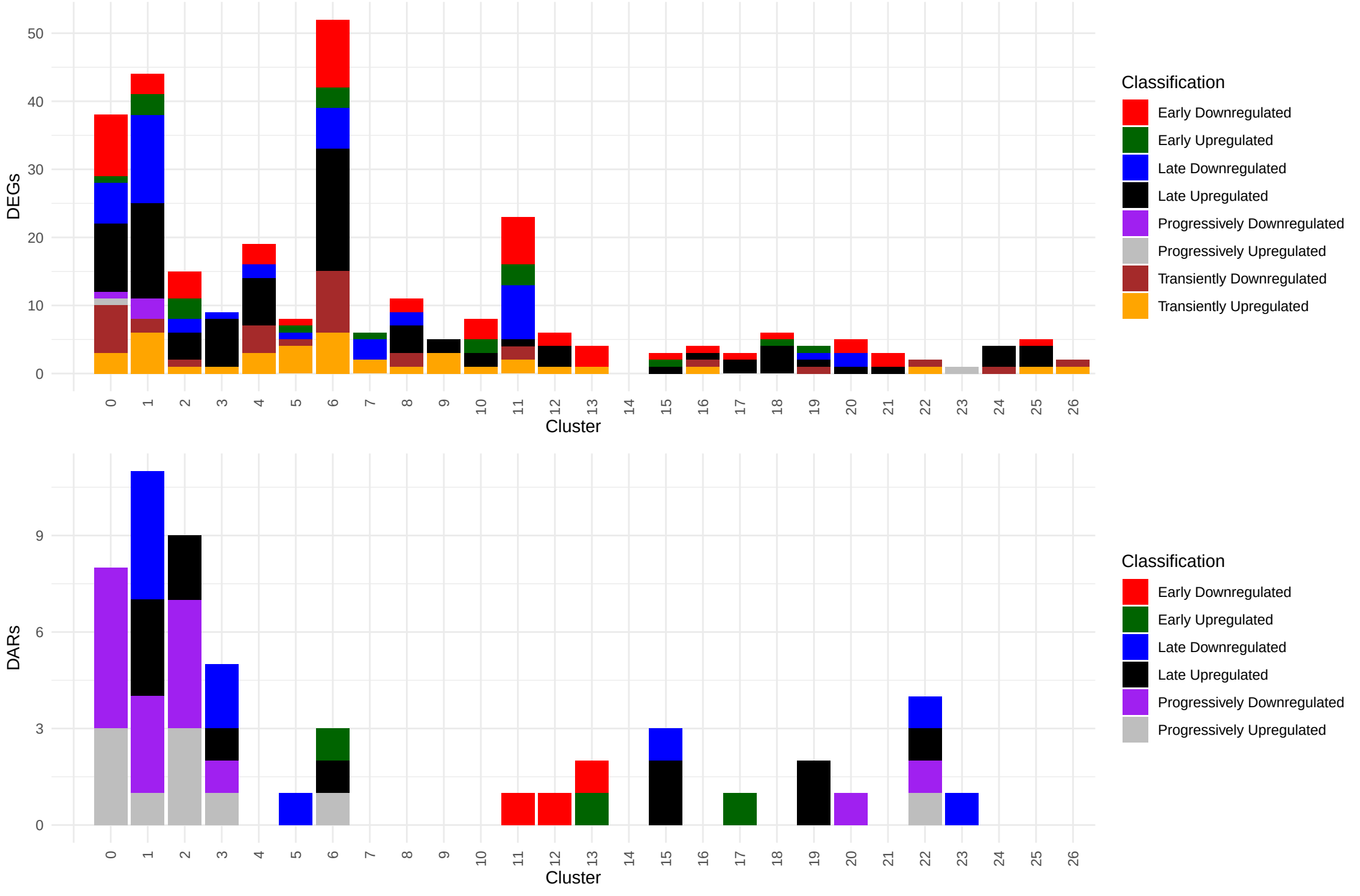

cyp19a1b | 1_RGC
p = 0.283
***
***
Normalized Expression
3
2
1
M
I
LI
NF
F
Phase

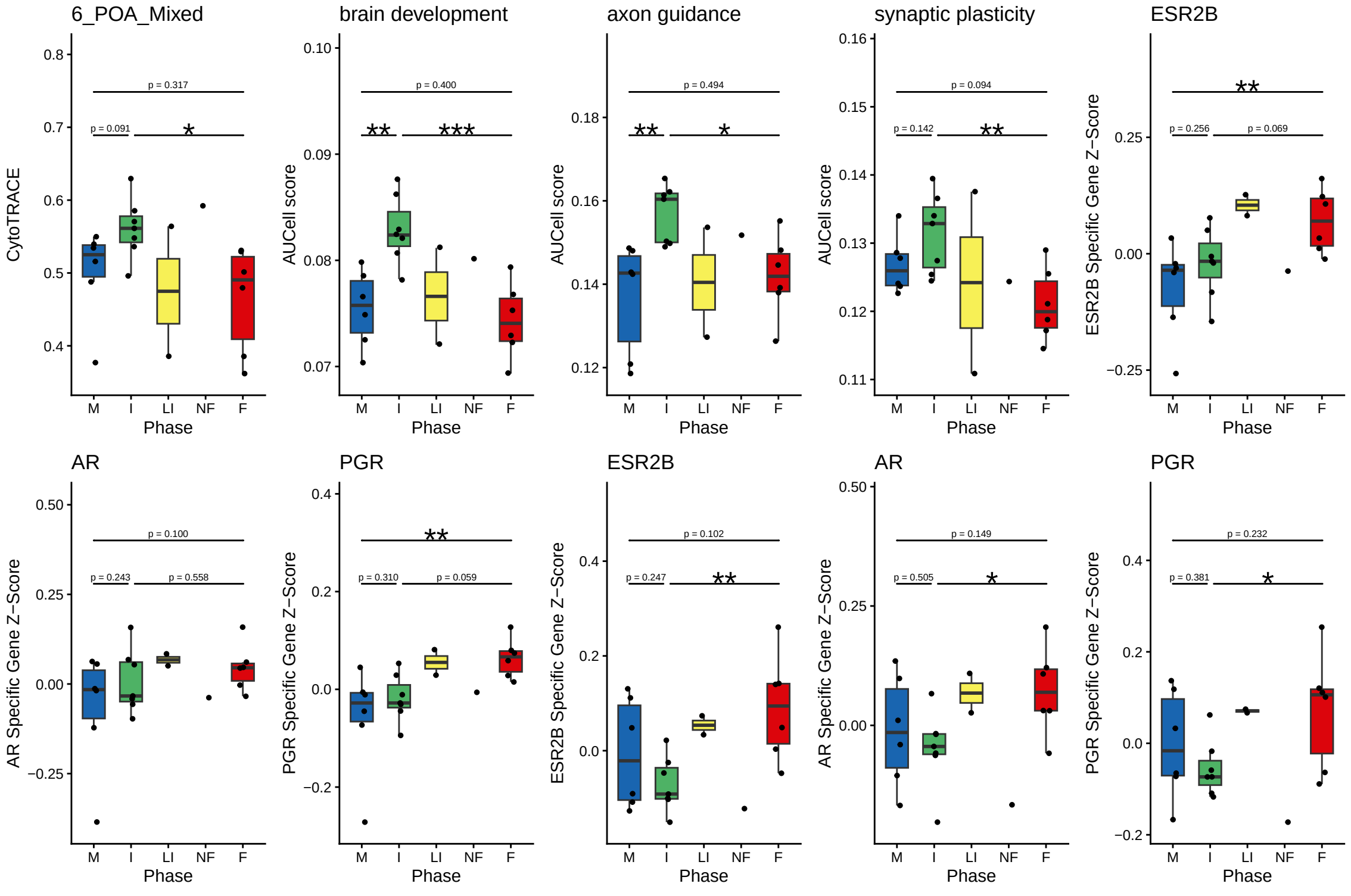

1_NSC
p = 0.356
**
*
% of 1_RGC
75%
50%
25%
0%
M
I
LI
NF
F
Phase

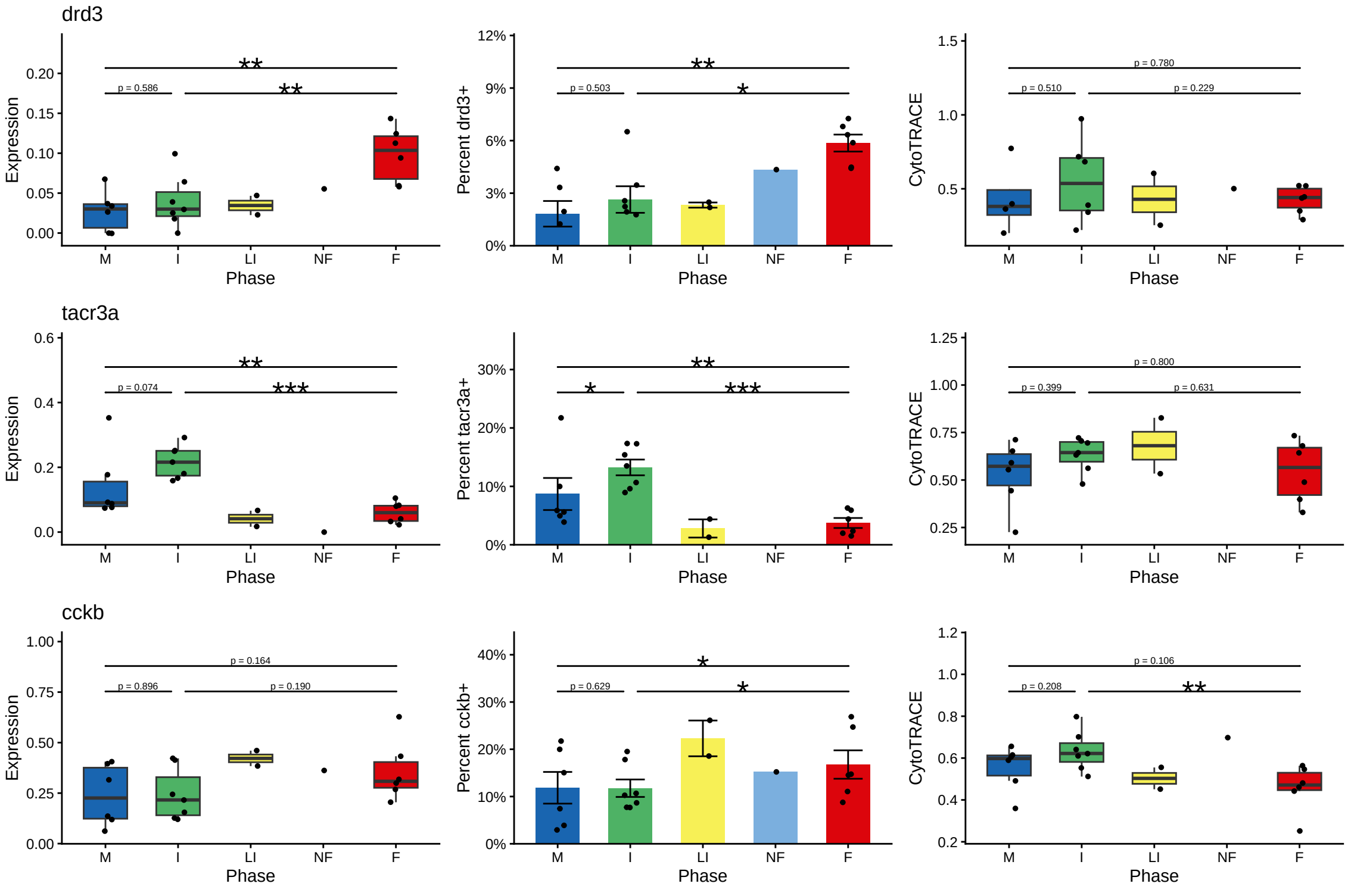

1_NP
**
*
***
% of 1_RGC
100%
75%
50%
25%
0%
M
I
LI
NF
F
Phase

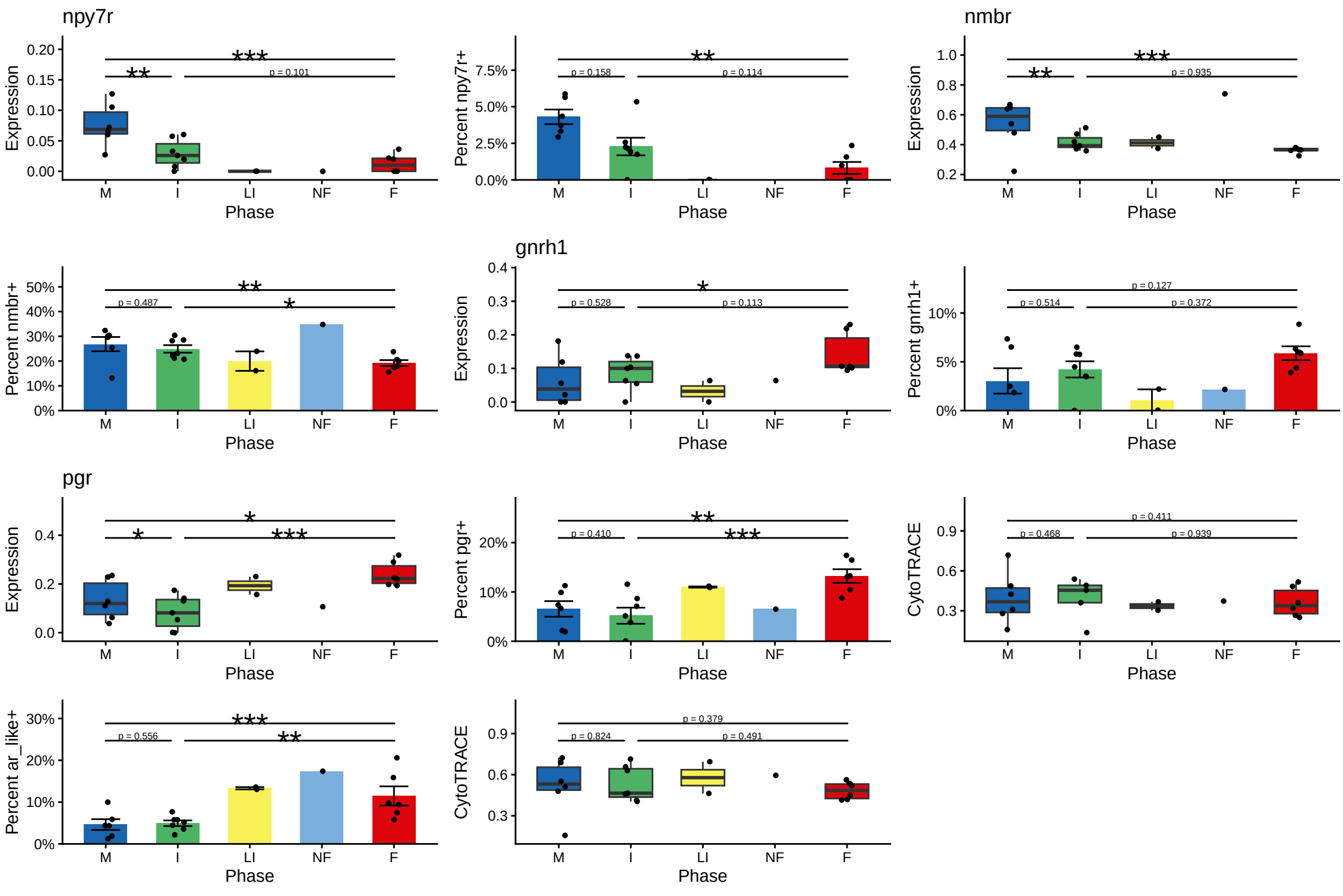

1_NP
**
p = 0.496
**
AUCell score
0.065
0.060
0.055
0.050
0.045
M
I
LI
NF
F
Phase

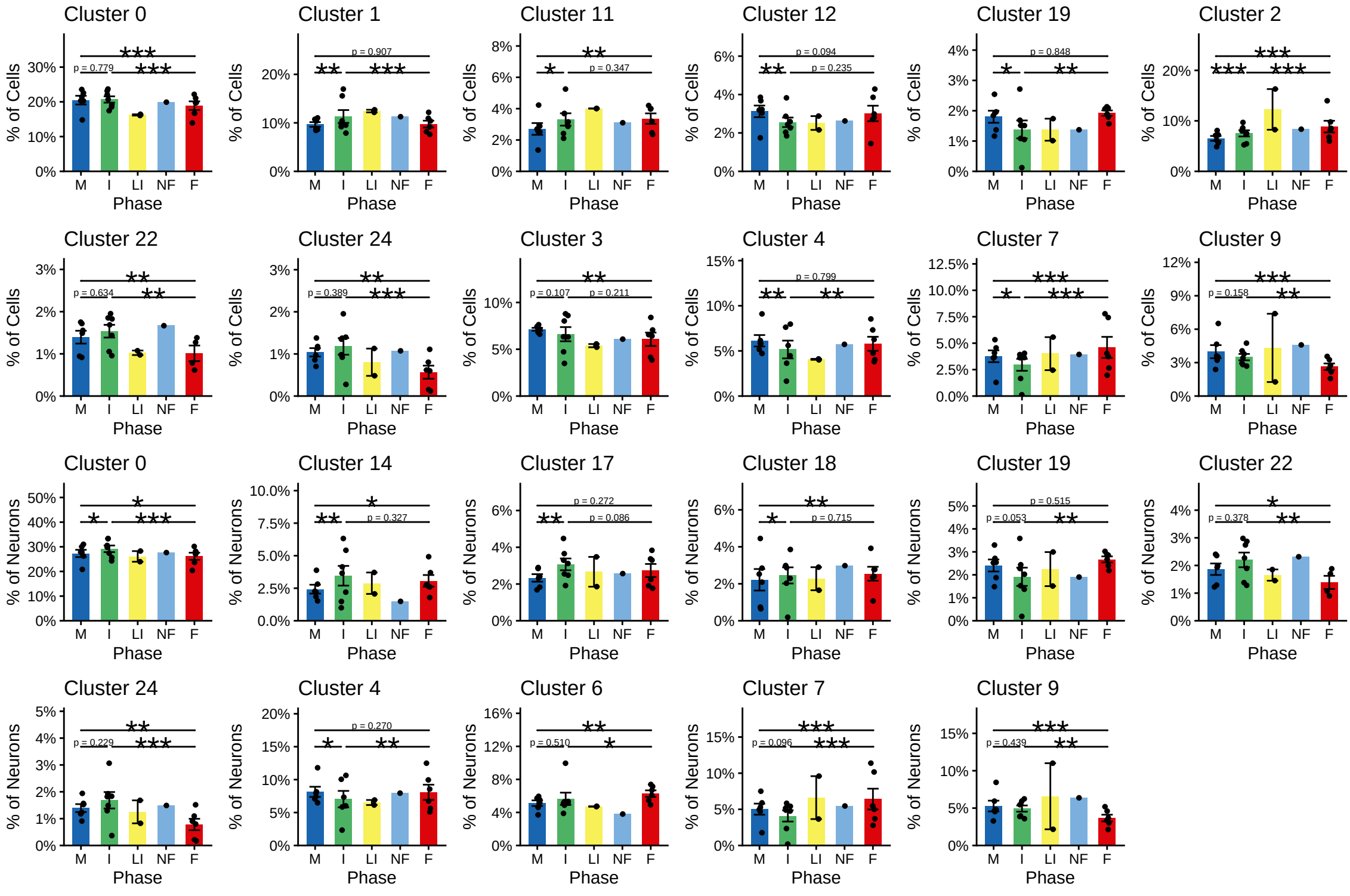

a
c
b
b
Mean CytoTRACE
1.00
0.75
0.50
0.25
0.00
1_0
1_1
1_2
1_3
Subcluster


DEGs
50
40
30
20
10
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26
Cluster
Classification
Early Downregulated
Early Upregulated
Late Downregulated
Late Upregulated
Progressively Downregulated
Progressively Upregulated
Transiently Downregulated
Transiently Upregulated
DARs
9
6
3
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26
Cluster
Classification
Early Downregulated
Early Upregulated
Late Downregulated
Late Upregulated
Progressively Downregulated
Progressively Upregulated


6_POA_Mixed
CytoTRACE
p = 0.317
p = 0.091
*
brain development
AUCell score
p = 0.400
**
***
axon guidance
AUCell score
p = 0.494
**
*
synaptic plasticity
AUCell score
p = 0.094
p = 0.142
**
ESR2B
ESR2B Specific Gene Z−Score
**
p = 0.256
p = 0.069
AR
AR Specific Gene Z−Score
p = 0.100
p = 0.243
p = 0.558
PGR
PGR Specific Gene Z−Score
**
p = 0.310
p = 0.059
ESR2B
ESR2B Specific Gene Z−Score
p = 0.102
p = 0.247
**
AR
AR Specific Gene Z−Score
p = 0.149
p = 0.505
*
PGR
PGR Specific Gene Z−Score
p = 0.232
p = 0.381
*
M
I
LI
NF
F
Phase


drd3
Expression
Percent drd3+
CytoTRACE
p = 0.586
p = 0.503
p = 0.780
p = 0.510
p = 0.229
tacr3a
Percent tacr3a+
p = 0.074
p = 0.800
p = 0.399
p = 0.631
cckb
Percent cckb+
p = 0.164
p = 0.896
p = 0.190
p = 0.629
p = 0.106
p = 0.208
M
I
LI
NF
F
Phase


npy7r
Expression
0.20
0.15
0.10
0.05
0.00
***
**
p = 0.101
M
I
LI
NF
F
Phase
Percent npy7r+
7.5%
5.0%
2.5%
0.0%
**
p = 0.158
p = 0.114
nmbr
Expression
1.0
0.8
0.6
0.4
0.2
***
**
p = 0.935
Percent nmbr+
50%
40%
30%
20%
10%
0%
**
*
p = 0.487
gnrh1
Expression
0.4
0.3
0.2
0.1
0.0
*
p = 0.528
p = 0.113
Percent gnrh1+
10%
5%
0%
p = 0.127
p = 0.514
p = 0.372
pgr
Expression
0.4
0.2
0.0
*
*
***
Percent pgr+
20%
10%
0%
**
***
p = 0.410
CytoTRACE
0.9
0.6
0.3
p = 0.411
p = 0.468
p = 0.939
Percent ar_like+
30%
20%
10%
0%
***
**
p = 0.556
CytoTRACE
0.9
0.6
0.3
p = 0.379
p = 0.824
p = 0.491


Cluster 0
Cluster 1
Cluster 11
Cluster 12
Cluster 19
Cluster 2
Cluster 22
Cluster 24
Cluster 3
Cluster 4
Cluster 7
Cluster 9
Cluster 0
Cluster 14
Cluster 17
Cluster 18
Cluster 19
Cluster 22
Cluster 24
Cluster 4
Cluster 6
Cluster 7
Cluster 9
% of Cells
% of Neurons
Phase
M
I
LI
NF
F